

Extracurriculars | EC

Summary

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# 1. CATEGORY THEORY IN CONTEXT - EMILY RIE

Notation:

- Morphism
- $\rightrightarrows$  Parallel pair of morphisms
- $\rightrightarrows$  Opposing pair of morphisms
- $\mapsto$  "maps to"
- $\rightsquigarrow$  "yields" / "leads to"
- $\dashrightarrow$  Implied existence of another morphism
- $=$  Genuine equality
- $:=$  Definitional equality
- $\cong$  Isomorphism
- $\simeq$  Weaker equivalence
- $\twoheadrightarrow$  Epimorphism
- $\hookrightarrow$  Inclusion/Embedding/"beinhaltet"/ $\subset$  ?

## 1.1. CHAPTER 1

### 1.1.1. Abstract and concrete categories

| Term                | Definition                                                                                      |
|---------------------|-------------------------------------------------------------------------------------------------|
| Commutative         | The order of the elements in an operation does not affect the result                            |
| Commutative diagram | Diagram where all directed paths with the same starting and ending points yield the same result |
| Natural             |                                                                                                 |
| Functor             |                                                                                                 |

Groups

| Term              | Definition                                                                                                                             |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Group             | A non-empty set with a binary operation satisfying associativity, identity and inverse<br>$\Leftrightarrow$ A groupoid with one object |
| Subgroup          |                                                                                                                                        |
| Group extension   |                                                                                                                                        |
| Abelian group     | A Group equipped with an operation that satisfies the commutative property                                                             |
| Abelian extension |                                                                                                                                        |
| Ring              |                                                                                                                                        |
| Quiver            | A directed graph defined by objects and morphisms in a category                                                                        |

Morphisms

| Term         | Definition                                                                            |
|--------------|---------------------------------------------------------------------------------------|
| Morphism     | A structure-preserving map between two objects within a category                      |
| Homomorphism | A morphism mapping between two algebraic structures while preserving their operations |

| <i>Term</i>  | <i>Definition</i>                                                                                                                                                                           |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Isomorphism  | A morphism $f : X \rightarrow Y$ for which there exists a morphism $g : Y \rightarrow X$ so that $gf = 1_X$ and $fg = 1_Y$<br>A morphism which is both a monomorphism and an epimorphism. * |
| Endomorphism | A morphism whose domain equals its codomain                                                                                                                                                 |
| Automorphism | An isomorphic endomorphism                                                                                                                                                                  |

## Categories

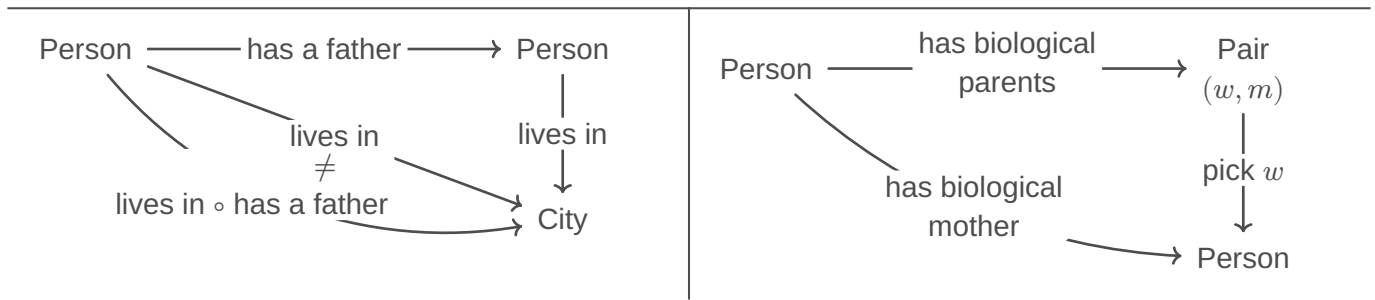
| <i>Term</i>            | <i>Definition</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Category               | Consists of objects $X, Y, Z, \dots$ and morphisms $f, g, h, \dots$ , such that: <ul style="list-style-type: none"> <li>– Each morphism has a specified domain and codomain among the collection of objects</li> <li>– Each object has a designated identity morphism</li> <li>– For any composable pair of morphisms <math>g, f</math> there exists a composite morphism <math>gf</math> whose domain is equal to the domain of <math>f</math> and whose codomain is equal to the codomain of <math>g</math></li> </ul> |
| Subcategory            | Subcollection of objects and morphisms of a category, such that it contains the domain and codomain of any morphism, the identity morphism of any object and the composite of any composable pair of morphisms in the subcategory                                                                                                                                                                                                                                                                                        |
| Small category         | A category is small if it has only a set's worth of arrows                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Locally small category | A category is locally small if between any pair of objects there is only a set's worth of morphisms                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Hom-set                | Set of arrows between a pair of fixed objects in a locally small category                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Discrete Category      | Every morphism is an identity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Slice category         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## Groupoids

| <i>Term</i>          | <i>Definition</i>                                                                                                                                                              |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Groupoid             | A category in which every morphism is an isomorphism $\Rightarrow$ has itself as its opposite category. Example: Discrete Category                                             |
| Fundamental groupoid | For any space $X$ , its fundamental groupoid $\Pi_1 X$ is a category whose objects are the points of $X$ and whose morphisms are endpoint-preserving homotopy classes of paths |
| Maximal groupoid     | The subcategory containing all of the objects and only those morphisms that are isomorphisms                                                                                   |
| Comonoid             |                                                                                                                                                                                |

## 1.1.1.1. Book club (04.03.26)

*Not commutative**Commutative*



**1.1.1.1.1. Let  $f : x \rightarrow y$ . Show that if  $g, h : y \rightrightarrows x$  so that  $gf = 1_x$  and  $fh = 1_y$  then  $g = h$  and  $f$  is an isomorphism.**

(i)  $a) f \circ h = 1_y$   
 $b) g \circ f = 1_x$   
 $g \stackrel{!}{=} h$

$$\begin{aligned} g &= g \circ 1_y \quad | \ a) \\ &= g \circ (f \circ h) \\ &= (g \circ f) \circ h \quad | \ b) \\ &= 1_x \circ h \\ &= h \end{aligned}$$

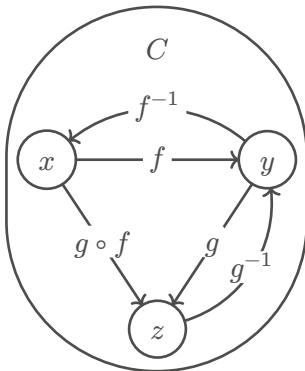
**1.1.1.1.2. Let  $C$  be a category. Show that the collection of isomorphisms in  $C$  defines a subcategory, the maximal groupoid inside  $C$**

There should exist a Category  $D$ , such that:

- Elements in  $D$ : The same as in  $C$
- Morphisms in  $D$ : The isomorphisms in  $C$

All identity morphisms can be taken into  $D$

Given isomorphisms  $f : x \rightarrow y, g : y \rightarrow z$ , show that  $g \circ f$  is isomorph.



$$\begin{aligned} (g \circ f)^{-1} &= f^{-1} \circ g^{-1} \\ \Leftrightarrow (g \circ f)^{-1} \circ (g \circ f) &= 1_x \\ \Leftrightarrow (g^{-1} \circ f^{-1}) \circ (g \circ f) &= 1_x \\ \Leftrightarrow g^{-1} \circ (f^{-1} \circ g) \circ f &= 1_x \\ \Leftrightarrow g^{-1} \circ 1_y \circ f &= 1_x \\ \Leftrightarrow g^{-1} \circ f &= 1_x \\ \Leftrightarrow 1_x &= 1_x \end{aligned}$$

$$\begin{aligned} (g \circ f) \circ (g \circ f)^{-1} &= 1_z \\ \Leftrightarrow g \circ f \circ f^{-1} \circ g^{-1} &= 1_z \\ \Leftrightarrow g \circ 1_y \circ g^{-1} &= 1_z \\ \Leftrightarrow g \circ g^{-1} &= 1_z \\ \Leftrightarrow 1_z &= 1_z \end{aligned}$$

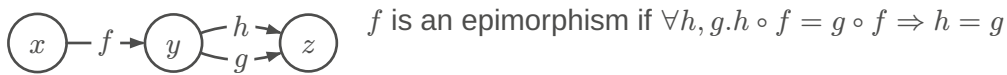
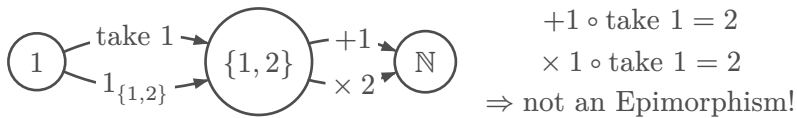
## 1.1.2. Duality

| Term              | Definition                                                                                                                                                                                                                                                                                       |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Duality           |                                                                                                                                                                                                                                                                                                  |
| Opposite Category | An opposite category of $C$ , called $C^{\text{op}}$ , has the same objects as $C$ and a morphism $f^{\text{op}}$ for each morphism $f$ in $C$ so that the domain of $f^{\text{op}}$ is defined to be the codomain of $f$ and the codomain of $f^{\text{op}}$ is defined to be the domain of $f$ |
| Composable        |                                                                                                                                                                                                                                                                                                  |
| Postcomposition   | $f_* : C(c, x) \rightarrow C(c, y), c \in C$                                                                                                                                                                                                                                                     |

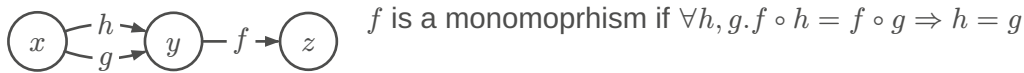
| Term                    | Definition                                                                                                     |
|-------------------------|----------------------------------------------------------------------------------------------------------------|
| Precomposition          | $f^* : \underbrace{C(y, c)}_{\text{Set of morphisms / "higher order morphisms"}} \rightarrow C(x, c), c \in C$ |
| Monomorphism            |                                                                                                                |
| Epimorphism             |                                                                                                                |
| Supremum                |                                                                                                                |
| Infimum                 |                                                                                                                |
| section/right inverse   |                                                                                                                |
| retraction/left inverse |                                                                                                                |
| retract                 |                                                                                                                |
| split morphism          |                                                                                                                |

### 1.1.2.1. Book club (11.03.26)

#### 1.1.2.1.1. Ex 1

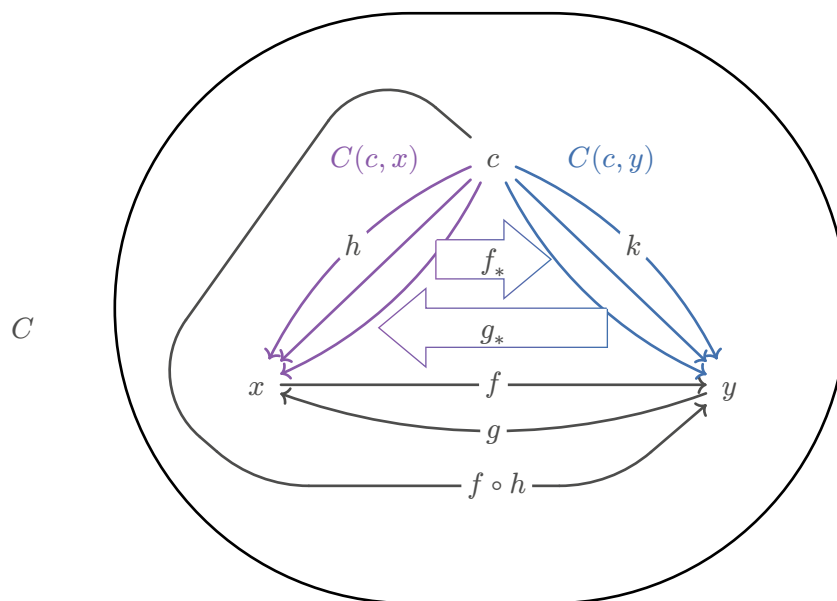


#### 1.1.2.1.2. Ex 2



### 1.1.2.2. Book club (18.03.26)

#### 1.1.2.2.1. Lemma 1.2.3



$$(i) \stackrel{!}{\Rightarrow} (ii)$$

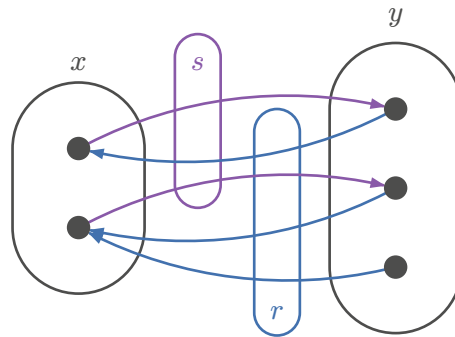
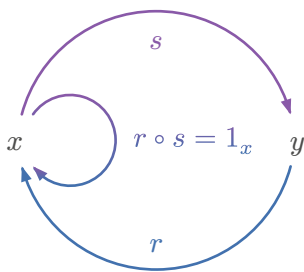
$$(1) \begin{cases} f_*(h) := f \circ h \\ f_* : h \mapsto f \circ h \end{cases}$$

$$(2) \begin{cases} g_*(k) := g \circ k \\ g_* : k \mapsto g \circ k \end{cases}$$

$$\left. \begin{aligned} g_*(f_*(h)) &\stackrel{(1)}{=} g_*(g \circ h) \stackrel{(2)}{=} \underbrace{g \circ f}_{(i) \Rightarrow 1_x} \circ h \stackrel{(i)}{=} h \\ f_*(g_*(k)) &\stackrel{(2)}{=} f_*(g \circ k) \stackrel{(1)}{=} \underbrace{f \circ g}_{(i) \Rightarrow 1_y} \circ k \stackrel{(i)}{=} k \end{aligned} \right\} g_* = (f_*)^{-1}$$

**TODO:**

$$(ii) \stackrel{!}{\Rightarrow} (i) \\ g_* = f_*^{-1}$$

**1.1.2.2.2. Split morphisms**

$$x \xrightarrow{s} y \xrightarrow{r} x \wedge r \circ s = 1_x$$

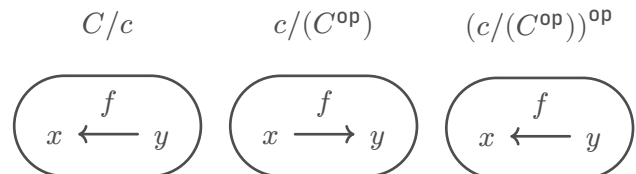
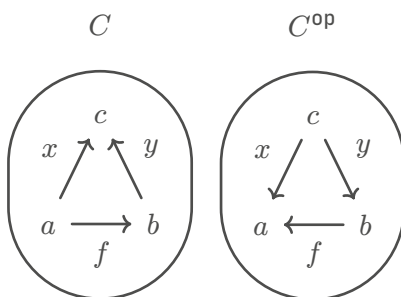
$\Rightarrow s$  is split monomorphism

(section/right inverse to  $r$ )

$\Rightarrow r$  is split epimorphism

(retraction/left inverse to  $s$ )

NOTE: that doesn't mean that  $s \circ r = 1_y$ .  
If so, then they would be isomorph.

**1.1.2.3. Book club (25.03.26)****1.2.ii**

$$(i) \text{ monic } f : x \rightarrowtail y \wedge g : y \rightarrowtail z \Rightarrow gf : x \rightarrowtail z$$

$$f : x \rightarrow y \Rightarrow h, k : w \rightrightarrows x, \underline{f \circ h = h \circ k} \stackrel{!}{\Rightarrow} h = k$$

$$(g \circ f) \circ h = (g \circ f) \circ k \stackrel{!}{\Rightarrow} h = k$$

$$\Leftrightarrow g \circ (f \circ h) = g \circ (f \circ k) \Rightarrow h = k$$

$$\Leftrightarrow g \circ (f \circ h) = g \circ (f \circ k) \Rightarrow h = k$$

$g$  monic

$$\Leftrightarrow f \circ h = f \circ k \Rightarrow h = k \quad f \text{ monic}$$

$$\Leftrightarrow h = k \Rightarrow k = k$$

(i') (dually)

(ii) monic  $g \circ f, f$

$$f \circ h = f \circ k \stackrel{!}{\Rightarrow} h = k$$

$$\Leftrightarrow g \circ (f \circ h) = g \circ (f \circ k) \Rightarrow h = k$$

$$\Leftrightarrow (g \circ f) \circ h = (g \circ f) \circ k \Rightarrow h = k \quad g \circ f \text{ monic}$$

$$\Leftrightarrow h = k \Rightarrow h = k$$

(ii') (dually)

#### 1.1.2.4. Book club (01.04.26)

1.2.iii

Fields are sets  $F$  with Operations  $+, -, \times, \div$

[https://abuseofnotation.github.io/category-theory-illustrated/02\\_category/](https://abuseofnotation.github.io/category-theory-illustrated/02_category/)

#### 1.1.2.5. Book club (15.04.26)

1.2.vi

$$y \xrightarrow{\quad} s \xrightarrow{\quad} r s \xrightarrow{1_x} k \xrightarrow{\quad} z$$

$$\overbrace{rs = 1_x}^{\text{Premisse}} \stackrel{?}{\Rightarrow} \overbrace{\forall h, k (hr = kr \Rightarrow h = k)}^{\text{To be shown}}$$

$$\forall h, k (hr = kr \Rightarrow hrs = krs \Rightarrow h1_x = k1_x \Rightarrow h = k)$$

Dual:

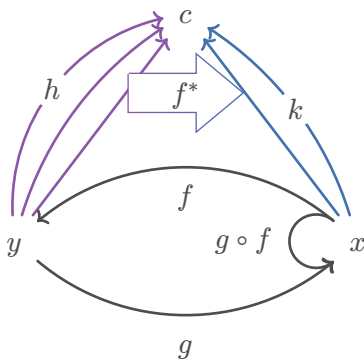
#### 1.1.2.6. Bool club (22.04.26)

1.2.v

(i)

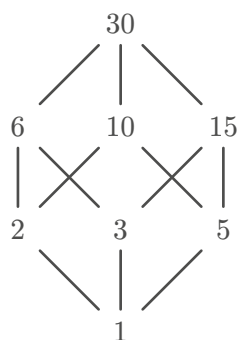
(ii) Let  $f^* : C(y, c) \rightarrow C(x, c)$  be surjective  $\forall c \in C$ . Show that only then  $f : x \rightarrow y$  is a split monomorphism in the category  $C$ .

$$k = h \circ f$$

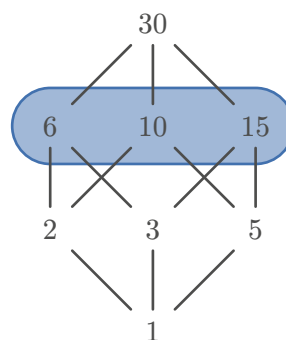


#### 1.1.2.7. Book club (29.04.26)

1.2.vii

**TODO:**

poset



Sup: 30, Inf: 1

**Supremum:** Smallest upper bound**Infimum:** Greatest lower bound $(\neg \leq) = (>)$ ,  $(\leq)^{\text{op}} = (\geq) \rightarrow$  opposite is not necessarily negation**1.1.2.8. Book club (06.05.26)**

“bag = multiset”

**1.1.2.9. Book club (20.05.26)**

Example 1.3.2. (i)

$$P : \begin{cases} \text{Set} \rightarrow \text{Set} \\ A \mapsto PA \end{cases}$$

$$f : A \rightarrow B$$

$$f_* : \begin{cases} PA \rightarrow PB \\ A' \mapsto \text{map } f A' \sim f(A') = \{f(a) | a \in A'\} \end{cases}$$

$$A' \subset A$$

$$1_{A*} : \begin{cases} PA \rightarrow PA \\ A' \mapsto \text{map } 1_{A'} = 1_{FA} \end{cases}$$



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## 2. OSDEV

useful links & cool projects

[OSDev wiki](#) , [osdever.net](#) , [it's good to be king](#) , [9front \(Plan 9\)](#) , [mirageOS](#) , [Uxn](#) , [seL4](#) ,

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### 2.1. KERNEL

#### 2.1.1. Memory

**Paging:** Memory gets subdivided into fixed-size blocks called **pages**.

**Segmentation:** Memory gets subdivided into variable-sized segments based on logical units such as functions, arrays, or data structures

**Virtual memory:** Exposing a virtual continuous block of memory to user-level applications which gets is to the underlying separate physical parts

**DMA:** Direct Memory Access

#### 2.1.2. Processes

##### 2.1.2.1. Inter Process Communication (IPC)

**Shared memory:** Processes can use shared memory for extracting information as a record from another process as well as for delivering any specific information to other processes

**Message passing:** Processes communicate by sending and receiving messages to exchange data. Message Passing can be achieved through different methods like Sockets, Message Queues or Pipes

**Gates:**

**IPC ports:**

##### 2.1.2.1.1. Multithreading/Synchronization

**Synchronization:** Maintaining the consistency of shared data even when multiple processes/threads are executed simultaneously.

**Critical region:** A region of memory shared by 2 or more processes

**Lock:** A variable whose value allows or denies the entrance to a critical region

**Semaphore:** A technique used to manage concurrent processes accessing a critical region by using a lock

**Counting Semaphore (Semaphore):** Allows access of a critical region to a group of processes

**Binary Semaphore (Mutex):** Allows access of a critical region to one process

**Busy waiting:** Continuously testing of a variable until some value appears

**Spinlock:** A lock which uses busy waiting

##### 2.1.2.1.2. Multitasking

**Round Robin:** Processes are put into a circular queue and executed sequentially

**Priority scheduling algorithm:** Each process is allocated a priority and the process with the highest priority is executed first

### 2.1.3. Syscalls

### 2.1.4. CPU

**Interrupts:** Signals that prompt the processor to temporarily halt its current tasks to address an event

**IRQ:** Interrupt Request

**ISR:** Interrupt Service Routine

#### 2.1.4.1. x86

**GDT:** Global Descriptor Table

**IDT:** Interrupt Descriptor Table

### 2.1.5. Filesystem

### 2.1.6. Hardware

#### 2.1.6.1. Device drivers